MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL,
LOW POWER, NAND GATES, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/20A, dated 20 November 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2, delete (including example of part number) and substitute:

"1.2 Part number. The part number shall be in accordance with MIL-M-38510."

* 1.2.3, Case outline letter B: Delete "1/8" and substitute "3/16".

PAGE 3

TABLE I, t_{PHL} test, maximum limits column: Device type 01, delete "140" and substitute "154"; device types 02 thru 06: Delete "90" and substitute "99".

* After 3.7(a), add the following new paragraph:

"3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 15 (see MIL-M-38510, Appendix E)."

PAGE 4

* TABLE II, first column, fourth test requirement: Delete "Groups B and C" and substitute "Groups C and D".

TABLE II, following the electrical test requirements table: Add "* PDA applies to subgroup 1 (see 4.3(h))."

* 4.2: In second sentence, delete "groups A, B, and C" and substitute "groups A, B, C, and D"; and delete third sentence in its entirety.

* 4.3(a), 4.3(b), 4.3(c), 4.3(e), and 4.3(g): Delete in their entirety.

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* 4.4.1, first sentence: Delete "consist of the test subgroups and LTPD values shown in" and substitute "be in accordance with".

* 4.4.2, 4.4.2(a), and 4.4.2(b), delete and substitute:

"4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of Method 5005 of MIL-STD-883."

* 4.4.3, delete and substitute:

"4.4.3 Groups C and D inspections. Groups C and D inspections shall be in accordance with tables III and IV of Method 5005 of MIL-STD-883 and as follows:".

* 4.4.3(b), line 1: Delete "Subgroups 7 and 8" and substitute "Subgroups 3 and 4".

* 4.4.3(c), 4.4.3(d), and 4.4.3(f): Delete in their entirety.

FSC 5962

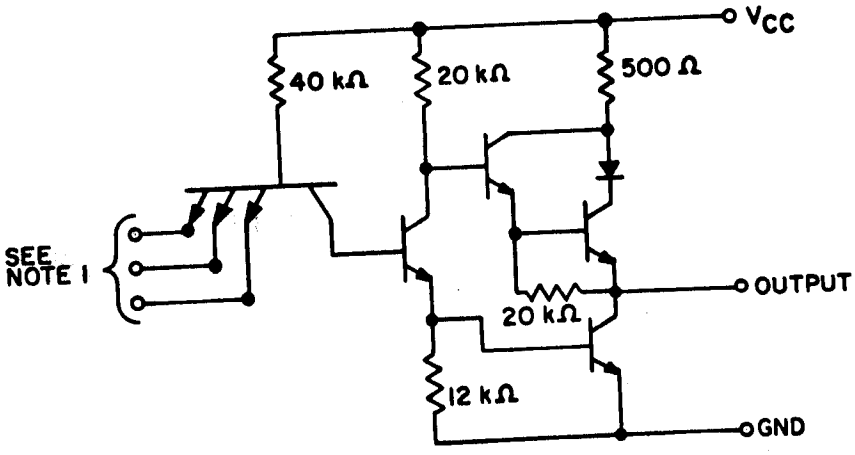
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PAGE 7

FIGURE 1, Logic diagram and terminal connections, device types 04 and 06: Delete "*" before device types 04 and 06 (top of page) and delete footnote "* Device type 06 in accordance with "Case C" only." following logic diagram and terminal connections for device types 04 and 06.

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FIGURE 3, Schematic circuits for device types 01 thru 05: Delete schematic B and substitute the following:



SCHEMATIC B

PAGE 12

FIGURE 5, Switching time test circuit, device under test: Delete "BUFFER" and substitute "GATE".

PAGES 12 AND 13

FIGURES 5 and 6, Switching time test circuits: Delete waveforms on both figures and substitute the following:

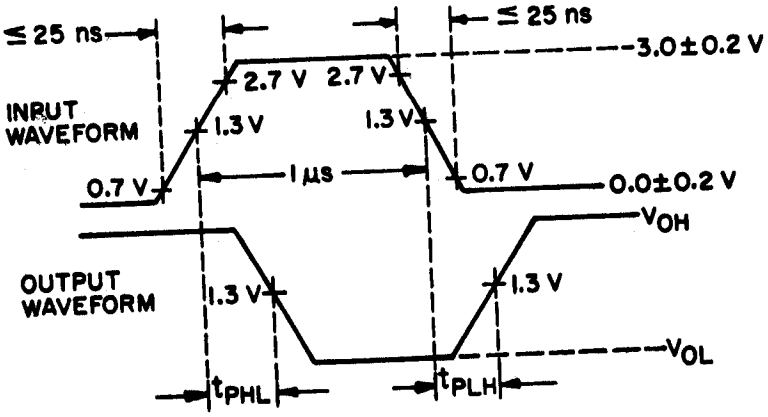


FIGURE 6, Switching time test circuit: Lead supplying V_{CC} to Gate under test should show electrically connected at intersection of V_{CC} and R_L.

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TABLE III, device type 01, t_{PHL}, test 37 CKT A, maximum limits column: Delete "90" and substitute "99".

TABLE III, device type 01, t_{PHL}, test 39 CKT A, maximum limits column: Delete "140" and substitute "154".

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TABLE III, device type 02, t_{PHL}, tests 39 CKT A and 40 CKT A, maximum limits column: Delete "60" and substitute "66".

TABLE III, device type 02, t_{PHL}, tests 43 CKT A and 44 CKT A, maximum limits column: Delete "90" and substitute "99".

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TABLE III, device type 03, t_{PHL}, tests 45 CKT A, 46 CKT A, and 47 CKT A, maximum limits column: Delete "60" and substitute "66".

TABLE III, device type 03, t_{PHL}, tests 51 CKT A, 52 CKT A, and 53 CKT A, maximum limits column: Delete "90" and substitute "99".

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TABLE III, device type 04, t_{PHL}, tests 43 CKT A, 44 CKT A, 45 CKT A, and 46 CKT A, maximum limits column: Delete "60" and substitute "66".

TABLE III, device type 04, t_{PHL}, tests 51 CKT A, 52 CKT A, 53 CKT A, and 54 CKT A, maximum limits column: Delete "90" and substitute "99".

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TABLE III, device type 05, t_{PHL}, tests 39 CKT A, 40 CKT A, 41 CKT A, 42 CKT A, 43 CKT A, and 44 CKT A, maximum limits column: Delete "60" and substitute "66".

TABLE III, device type 05, t_{PHL}, tests 51 CKT A, 52 CKT A, 53 CKT A, 54 CKT A, 55 CKT A, and 56 CKT A, maximum limits column: Delete "90" and substitute "99".

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TABLE III, device type 06: Delete and substitute TABLE III, device type 06 as shown on page 4 of this amendment.

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TABLE III, device type 06 - Continued: Delete and substitute TABLE III, device type 06 - Continued as shown on page 5 of this amendment.

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	1	2	3	14	6	4	5	8	9	10	7	12	13	11		Min	Max	Unit
			Test No.	1A	1B	1Y	V_{CC}	2Y	2A	2B	3Y	3A	3B	GND	4A	4B	4Y				
1 T_A=25°C	V_{OL}	3007	1	2.0 v	2.0 v	2 mA	4.5 v		5.5 v	5.5 v		5.5 v	5.5 v	GND	5.5 v	5.5 v		1Y		0.3	V
			2	5.5 v	5.5 v			2 mA	2.0 v	2.0 v		5.5 v	5.5 v					2Y			
			3						5.5 v	5.5 v	2 mA	2.0 v	2.0 v					3Y			
			4									5.5 v	5.5 v		2.0 v	2.0 v	2 mA	4Y			
	I_{CEX}		5	0.6 v	4.5 v	5.5 v									5.5 v	5.5 v		1Y		50	µA
			6	4.5 v	0.6 v	5.5 v												1Y			
			7	5.5 v	5.5 v			5.5 v	0.6 v	4.5 v								2Y			
			8					5.5 v	4.5 v	0.6 v								2Y			
			9						5.5 v		5.5 v	0.6 v	4.5 v					3Y			
			10								5.5 v	4.5 v	0.6 v					3Y			
			11												0.6 v	4.5 v	5.5 v	4Y			
			12												4.5 v	0.6 v	5.5 v	4Y			
	I_{IH1}	3010	13	2.4 v	GND		5.5 v		GND	GND		GND	GND		GND	GND		1A		10	
			14	GND	2.4 v				GND									1B			
			15		GND				2.4 v									2A			
			16						GND	2.4 v								2B			
			17							GND		2.4 v						3A			
			18									GND	2.4 v					3B			
			19										GND		2.4 v			4A			
			20										GND		GND	2.4 v		4B			
	I_{IH2}		21	5.5 v	GND					GND			GND			GND		1A		100	
			22	GND	5.5 v					GND								1B			
			23		GND				5.5 v	GND								2A			
			24						GND	5.5 v								2B			
			25							GND		5.5 v						3A			
			26									GND	5.5 v					3B			
			27										GND		5.5 v			4A			
			28										GND		GND	5.5 v		4B			
	I_{IL}	3009	29	0.3 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v		5.5 v			1A	-60	-180	
			30	5.5 v	0.3 v				5.5 v									1B			
			31		5.5 v				0.3 v	0.3 v								2A			
			32						5.5 v	5.5 v								2B			
			33									0.3 v						3A			
			34									5.5 v	0.3 v					3B			
			35										5.5 v		0.3 v			4A			
			36												5.5 v	0.3 v		4B			
	I_{CCL}	3005	37												5.5 v	5.5 v		V_{CC}		2.04	mA
	I_{CCH}	3005	38	GND	GND				GND	GND		GND	GND		GND	GND		V_{CC}		0.80	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C.																				

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	1	2	3	14	6	4	5	8	9	10	7	12	13	11		Min	Max	Unit
			Test No.	1A	1B	1Y	V_{CC}	2Y	2A	2B	3Y	3A	3B	GND	4A	4B	4Y				
9 T_A=25°C	t_{PHL}	3003 (Fig. 6)	39 40 41 42	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	OUT	IN	2.4 v	GND	IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	3	66	ns
	t_{PLH}		43 44 45 46	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		90	
10 T_A=125°C	t_{PHL}		47 48 49 50	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		99	
	t_{PLH}		51 52 53 54	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		140	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55°C.																				

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Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 70, 99
DSA - ES

User activities:
Army - SM, MU, WC
Navy - AS, OS, CG, MC
Air Force - 13, 15, 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

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